



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Machine Design) (CBCS) Degree Examinations

May - June 2024

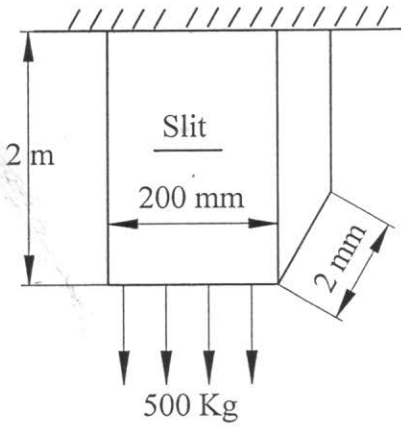
22MMD223 – FRACTURE MECHANICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

❖ Use of **Fracture Mechanics Data Hand Book** is permitted.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define fracture. List the use of fracture mechanics in modern engineering design.	6	L*2	1	1
	b) Derive an expression for strength of an ideal material.	14	L4	1	1
2.	a) Define stress intensity factor. Provided SIF solution to any five cracked body problems.	8	L2	1	1
	b) Calculate the fracture strength of a thin shell steel plate, containing a central elliptical crack of 2.5 mm size loaded perpendicular to the plate of the crack. Take $E = 207 \text{ GPa}$ and $\gamma_s = 2 \text{ J/m}^2$	6	L2	2	2
	c) List the various NDT methods used in fracture mechanics. Explain any one NDT methods.	6	L2	2	1
3.	a) Differentiate clearly between ductile fracture and brittle fracture.	6	L2	2	2
	b) Fig (Q.2c) shows a sheet of glass of designated dimensions containing a small (Slit) crack. A mass of 500 kg induces a tensile stress in the sheet and is assumed here that any effect of the mass the glass is included in 500 kg value. If $E = 60 \text{ GPa}$, $\gamma_s = 0.5 \text{ J/m}^2$, $\nu = 0.25$ find the minimum length of slit.				
		6	L2	2	2
	Fig.(Q 2c)				
	c) A flat plate with a through thickness crack subjected to 100 MPa, tensile test has a fracture toughness of $50 \text{ MPa}\sqrt{\text{m}}$. Determine critical crack length in a plate. Also find critical energy release rate for the plate (take $E = 207 \text{ GPa}$).	8	L3	2	2
Unit – II					
4.	a) Discuss the effect of thickness on fracture toughness.	8	L4	3	2
	b) Explain: (i). Dugdale's model for plastic zone size. (ii). Draw the shapes and size of the plastic zone for plane stress and strain according Von-Misses and Tresca yield criteria.	12	L3	3	2

P.T.O

5. a) A thick center crack plate of high strength aluminum alloy is 200 mm wide and contains a crack of length 80 mm. if it fails at an applied stress of 100 MPa, what is the fracture toughness of the alloy? what value of applied stress would produce fracture for the same length of crack in,
 i) Infinite body. 8 L4 3 2
 ii) 120 mm wide plate.
- b) Give difference between J – integral and energy release rate [G]. 6 L2 4 1
- c) Explain R curve of ductile and brittle materials with the help of a graph explain stable and unstable crack growth. 6 L2 4 2
6. a) Explain the principles of crack arrest, with neat sketch. Explain any two techniques for arresting the propagating crack. 10 L2 4 1
- b) A 4340 steel has $C = 1.2 \times 10^{-11} \frac{m}{cycle}$, $\sqrt{m} = 2.8$, $\sigma_{ys} = 1000 \text{ MPa}$;
 $\sigma_u = 1050 \text{ MPa}$ & $K_{IC} = 185 \text{ MPa}$. and is loaded cyclically between zero to 250 MPa. If $a_i = 0.0001 \text{ m}$; $f(c) = 1$, determine, 10 L4 4 2
- i. Maximum crack length after 150×10^3 cycles.
 ii. Number of cycles to failure.

Unit – III

7. a) Explain Analytical, experimental and numerical methods used to determine stress intensity factors. 10 L2 5 1
- b) Write short note on:
 i). Applicability 10 L2 5 1
 ii). Non- linearity
 iii). Estimation of SIT
8. a) Explain the fracture toughness (K_{IC}) test procedure for a SENB specimen. 10 L2 5 1
- b) The fracture toughness of a material is $55 \text{ MPa}\sqrt{m}$, yield stress is 515 MPa and modulus of elasticity is 70 GPa . The plate has a through thickness-central crack of $2a = 15 \text{ mm}$. Calculate the plastic zone size at the fracture and the fracture stress. Assume plane stress state. 10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
